

Instructor: Date:

Learner: Regional Branch:

Program: Center:

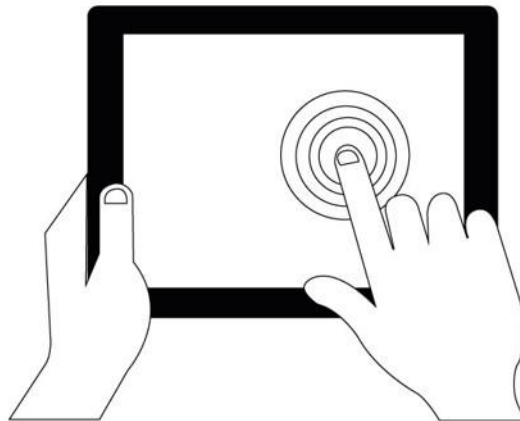
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

An Introduction to Software Testing



Learning Outcome:

Implement and document the software tests, applying trial and error, based on the designed plan and procedures established by the company.

Applies the different types of software testing to the IT solution.

Ask your partner:

- Do you consider important to check the quality of a product before releasing it to the market? Why?
- What would you do if defective software damaged your personal computer?

Can Do Statements:

- I can explain what software testing is.
- I can describe the two basic software testing.
- I can establish differences between the white box and the black box testing.



Let's Explore!

Samuel helps Maggie out with an assignment about software testing.



Instructions: Read the text below.

- Maggie:** Hey, Sam! Would you give me a hand with some concepts from my systems class? There are some things I don't understand very well. I know you're keen on computer programming.
- Samuel:** Yeah! I'm not that bad... Tell me how I can help you.
- Maggie:** Well, I don't really understand software testing.
- Samuel:** Just keep in mind that If you don't want to implement bad software, you will have to test it to avoid bugs popping up. That's why you have to run tests on a software system before anyone uses it. It basically confronts the given input and the desired output.
- Maggie:** So, if I don't want my software system to fail, I will have to examine it thoroughly. Got it! I don't understand either the two basic software systems. The two basic software systems: the black box and the white box.
- Samuel:** To begin with, software testing should occur at different stages of development. A black box testing is focused on the input and output of the system. It totally misses the inner part of the mechanism. A white box testing is, on the contrary, focused on the internal logic of the mechanism. That's why it's called white box or glass box testing because it allows you to see through the system.
- Maggie:** Ok, but what are the main differences that exist between black and white box testing?
- Samuel:** Well, the black box is mostly used for validation and the white box is used for verification. While the black box is functional testing, the white box is structural. With that being said, you should remember that if you want to get a better understanding of the inner mechanism of the system, you don't have to go through the black box but the white box one.
- Maggie:** Wow! If you weren't my friend, I wouldn't know what to do. The test that I have tomorrow will be easier to pass.
- Samuel:** I'm sure you will pass it. Is there anything else I can help you with?





Let's Practice!

If you don't want any inconveniences in your system you'll need to check the two basic software testing.

Instructions: Match the sentences from Column A with Column B.

COLUMN A

1. If you want to implement a black box testing,
2. If you don't want your software system to fail,
3. If I needed to see through the system,
4. If you use a black box testing you,
5. If you use a white box testing,

COLUMN A

- a. I won't use a black box testing.
- b. you won't consider the inner part of the mechanism.
- c. you won't focus on the input and output of the system but only on the internal logic of the mechanism.
- d. you will have to examine it thoroughly.
- e. you won't focus on the internal logic of the mechanism but only on the input and output of the system.



Let's Practice!

Remember a couple of differences between the white box and the black box testing.

Instructions: Underline all the subject and object relative clauses.

Maggie: Ok, but what are the main differences that exist between black and white box testing?

Samuel: Well, the black box that is mostly used for validation is a pretty useful one. The white box, which is used for verification, it's also known as a glass box testing. There's a difference between the white box and the black box testing the black box which is functional testing and the white box that is structural testing.



Extension Activity

Designing a comparative chart.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about black box and white box testing methods. Extract the most important aspects about them.
- Create a comparative chart explaining the advantages and disadvantages of both methods. Include your own ideas using the grammar you studied.
- Exchange your chart with a partner and then find commonalities between both charts.

Self-Assessment

Yes

No

- I look for more information about black box and white box testing methods. I extract the most important aspects about them. ☐ ☐
- I create a comparative chart explaining the advantages and disadvantages of both methods. I include my own ideas using the grammar studied. ☐ ☐
- I exchange my chart with a partner and then I find commonalities between both charts. ☐ ☐



Target Words

- **be keen on** (expression): being enthusiastic on doing something or related to a topic.
e.g. I know you're keen on computer programming.
- **bug** (noun): in informatics referring to an error in programming.
e.g. Just keep in mind that If you don't want to implement bad software, you will have to test it to avoid bugs popping up.
- **run a test** (expression): to perform a test.
e.g. That's why you have to run tests on a software system before anyone uses it.
- **thoroughly** (adverb): to do something in detail.
e.g. So, if I don't want my software system to fail, I will have to examine it thoroughly.
- **be focused on** (expression): to concentrate on something specific.
e.g. A white box testing is, on the contrary, focused on the internal logic of the mechanism.
- **inner** (adjective): something found in the core, center or inside.
e.g. It totally misses the inner part of the mechanism.



Learning Points

Conditional Sentences with If Clauses-Negative Form

A **conditional sentence** states something that happens, will happen, or could happen but depends or is closely linked to a condition.

Conditional clauses talk about situations that need a condition to be true. The final result depends on the condition to be real. There are different types of **conditional sentences** with different degrees of grammatical structures.

The usual **conditional sentence** is composed of 2 clauses: the *if clause* that explains the **condition**, and the other clause that presents the **result** or consequence.

Any of the two clauses can be affirmative, negative or even mixed. You just have to be clear about the meaning and the idea you want to express.

Negative + negative form:

If you **don't** save enough money, you **won't** be able to buy a laptop.

Affirmative + negative form:

If you **decide to save** your lunch money, you **won't** be able to spend time eating with your friends.

Negative + affirmative form:

If Theresa **doesn't** save enough money for her scholarship, she **will** feel really sad.

Examples

- *So, if I don't want my software system to fail, I will have to examine it thoroughly.*
- *Just keep in mind that If you don't want to implement bad software, you will have to test it to avoid bugs popping up.*
- *If you weren't my friend, I wouldn't know what to do.*
- *With that being said, you should remember that if you want to get a better understanding of the inner mechanism of the system, you don't have to go through the black box but the white box one.*

What is software testing?

Software testing is a very important step in the development of software systems for different applications.





Learning Points

Any software system has to go through a series of tests that help to spot mistakes during different stages of development. Such mistakes are called bugs and they represent a great hazard to the correct functioning of the software. There have been cases where such failures have caused deaths, for instance, when airplanes and medical appliances presented malfunctioning during important procedures.

They also represent loss worth millions of dollars and that is why software developers emphasize on software testing. There are many types of testing but the two most common are the black box and the white box testing.

The **black box** test assesses the agreement between input (given instructions for a system) and output (the expected result) which focuses on the outer elements of software.

The **white box** or glass box test focuses on the inside of the mechanism. It assesses internal logic on any program. It is called white or glass box because it allows seeing the insides of a program.

Overall, **software testing** is the evaluation of a software system that assesses the quality of the product in different kinds of performance.